CSES Problem Set - Required Algorithms Guide

General & Introduction

(No algorithms assigned to UI elements like Create new account, Statistics)

Introductory Problems

- **Weird Algorithm** — Simulation / Recursion
- **Missing Number** — Math (Sum formula) / Bitwise XOR
- **Repetitions** — Linear Scan / Greedy
- **Increasing Array** — Greedy
- **Permutations** — Constructive Algorithm / Math
- **Number Spiral** — Math / Geometry
- **Two Knights —** Combinatorics / Math
- **Two Sets** — Math / Constructive Algorithm
- **Bit Strings —** Fast Exponentiation / Modular Arithmetic
- **Trailing Zeros** — Math (Prime Factorization)
- **Coin Piles —** Math / Modular Arithmetic
- **Palindrome Reorder** — String Manipulation / Greedy / Hashing (Character counting)
- **Gray Code —** Bitwise Operations / Recursion
- **Tower of Hanoi** — Recursion
- **Creating Strings —** Backtracking / Next Permutation (Combinatorics)
- **Apple Division** — Complete Search / Bitmask / Backtracking
- **Chessboard and Queens —** Backtracking / Pruning
- **Raab Game I** — Math / Ad-hoc
- **Mex Grid Construction —** Constructive Algorithm
- **Knight Moves Grid** — BFS / Backtracking
- **Grid Coloring I —** Constructive Algorithm / Math
- **Digit Queries** — Binary Search / Math
- **String Reorder** — Greedy / Priority Queue
- **Grid Path Description** — DFS / Backtracking / Pruning

Sorting and Searching

- **Distinct Numbers** — Sorting / Set
- **Apartments** — Sorting / Two Pointers / Greedy
- **Ferris Wheel** — Sorting / Two Pointers / Greedy
- **Concert Tickets** — Sorting / Binary Search (Multiset)
- **Restaurant Customers** — Sweep Line / Sorting / Prefix Sums
- **Movie Festival** — Sorting / Greedy (Interval Scheduling)
- **Sum of Two Values** — Sorting / Two Pointers / Hash Map
- **Maximum Subarray Sum** — Kadane's Algorithm
- **Stick Lengths** — Sorting / Median Finding
- **Missing Coin Sum** — Sorting / Greedy
- **Collecting Numbers** — Array Indexing / Greedy
- **Collecting Numbers II** — Array Indexing / Simulation
- **Playlist** — Two Pointers / Sliding Window / Set
- **Towers** — Greedy / Binary Search (Multiset) / LIS variant
- **Traffic Lights** — Set / Multiset / BST (Binary Search Tree)
- **Distinct Values Subarrays** — Two Pointers / Hash Map
- **Distinct Values Subsequences** — Dynamic Programming
- **Josephus Problem I** — Simulation / Queue / Linked List
- **Josephus Problem II** — PBDS (Order Statistic Tree) / Segment Tree / Fenwick Tree
- **Nested Ranges Check** — Sorting / Sweep Line
- **Nested Ranges Count** — Sorting / Fenwick Tree (BIT) / Segment Tree
- **Room Allocation** — Sorting / Sweep Line / Priority Queue
- **Factory Machines** — Binary Search on Answer
- **Tasks and Deadlines** — Sorting / Greedy
- **Reading Books** — Sorting / Greedy / Math
- **Sum of Three Values —** Sorting / Two Pointers
- **Sum of Four Values** — Sorting / Hash Map / Two Pointers
- **Nearest Smaller Values —** Monotonic Stack
- **Subarray Sums I** — Two Pointers / Sliding Window
- **Subarray Sums II —** Prefix Sums / Hash Map

- **Subarray Divisibility** — Prefix Sums / Hash Map / Modular Arithmetic
- **Distinct Values Subarrays II** — Two Pointers / Map
- **Array Division** — Binary Search on Answer / Greedy
- **Movie Festival II** — Sorting / Greedy / Multiset
- **Maximum Subarray Sum II** — Sliding Window / Monotonic Queue / Deque

Dynamic Programming

- **Dice Combinations** — 1D Dynamic Programming
- **Minimizing Coins —** Unbounded Knapsack DP
- **Coin Combinations I** — Unbounded Knapsack DP (Ordered)
- **Coin Combinations II —** Unbounded Knapsack DP (Unordered)
- **Removing Digits** — 1D Dynamic Programming / Greedy
- **Grid Paths I —** 2D Dynamic Programming
- **Book Shop** — 0/1 Knapsack DP
- **Array Description —** 2D Dynamic Programming
- **Counting Towers** — 2D Dynamic Programming / State Machine DP
- **Edit Distance —** 2D Dynamic Programming (String Alignment)
- **Longest Common Subsequence** — 2D Dynamic Programming
- **Rectangle Cutting —** 2D Dynamic Programming
- **Minimal Grid Path** — 2D Dynamic Programming
- **Money Sums —** 0/1 Knapsack DP / Bitset DP
- **Removal Game** — Interval DP / Game Theory DP
- **Two Sets II** — 0/1 Knapsack DP
- **Mountain Range** — Combinatorics / Catalan Numbers / DP
- **Increasing Subsequence** — DP with Binary Search / Segment Tree / Patience Sorting
- **Projects** — Coordinate Compression / DP / Binary Search / Segment Tree
- **Elevator Rides** — Bitmask Dynamic Programming
- **Counting Tilings** — DP on Broken Profile / Bitmask DP
- **Counting Numbers** — Digit Dynamic Programming
- **Increasing Subsequence II** — DP + Fenwick Tree / Segment Tree

Graph Algorithms

- **Counting Rooms** — DFS / BFS / Connected Components
- **Labyrinth** — BFS
- **Building Roads** — DFS / BFS / DSU (Disjoint Set Union)
- **Message Route** — BFS
- **Building Teams** — Bipartite Graph Checking / BFS / DFS
- **Round Trip** — Cycle Detection (DFS)
- **Monsters** — Multi-source BFS
- **Shortest Routes I** — Dijkstra's Algorithm
- **Shortest Routes II** — Floyd-Warshall Algorithm
- **High Score** — Bellman-Ford Algorithm
- **Flight Discount** — Dijkstra's Algorithm with State / DP on Graphs
- **Cycle Finding** — Bellman-Ford Algorithm
- **Flight Routes** — Dijkstra's Algorithm (k-th shortest path)
- **Round Trip II** — Cycle Detection in Directed Graph (DFS)
- **Course Schedule** — Topological Sort (Kahn's or DFS)
- **Longest Flight Route** — Topological Sort / DP on DAG
- **Game Routes** — Topological Sort / DP on DAG
- **Investigation** — Dijkstra's Algorithm / DP on DAG
- **Planets Queries I** — Binary Lifting / Successor Graphs
- **Planets Queries II** — Successor Graphs / Cycle Detection
- **Planets Cycles** — Successor Graphs / Cycle Detection
- **Road Reparation** — Minimum Spanning Tree (Kruskal's or Prim's Algorithm)
- **Road Construction** — DSU (Disjoint Set Union)
- **Flight Routes Check** — Strongly Connected Components (Kosaraju's or Tarjan's)
- **Planets and Kingdoms** — Strongly Connected Components (Kosaraju's or Tarjan's)
- **Giant Pizza** — 2-SAT
- **Coin Collector —** SCC + Condensation Graph + DP on DAG
- **Mail Delivery** — Eulerian Circuit (Hierholzer's Algorithm)
- **De Bruijn Sequence —** Eulerian Circuit
- **Teleporters Path** — Eulerian Path
- **Hamiltonian Flights —** Bitmask DP on Graph

- **Knight's Tour** — Warnsdorff's Heuristic / Backtracking
- **Download Speed** — Maximum Flow (Edmonds-Karp / Dinic's Algorithm)
- **Police Chase** — Minimum Cut (Max Flow Min Cut Theorem)
- **School Dance** — Bipartite Matching (Max Flow / Hopcroft-Karp Algorithm)
- **Distinct Routes** — Maximum Flow + Path Retrieval

Range Queries

- **Static Range Sum Queries** — Prefix Sums
- **Static Range Minimum Queries —** Sparse Table / Segment Tree
- **Dynamic Range Sum Queries** — Fenwick Tree (BIT) / Segment Tree
- **Dynamic Range Minimum Queries —** Segment Tree
- **Range Xor Queries** — Prefix XORs
- **Range Update Queries —** Difference Array + Prefix Sums / Fenwick Tree / Lazy Segment Tree
- **Forest Queries** — 2D Prefix Sums
- **Hotel Queries** — Segment Tree (Binary Search on Segment Tree)
- **List Removals** — Segment Tree / Fenwick Tree / PBDS
- **Salary Queries** — Coordinate Compression + Fenwick Tree / PBDS / Segment Tree
- **Prefix Sum Queries** — Lazy Segment Tree
- **Pizzeria Queries** — Segment Tree
- **Visible Buildings Queries** — Segment Tree / Monotonic Stack
- **Range Interval Queries** — Mo's Algorithm / Sweep Line + Segment Tree
- **Subarray Sum Queries** — Segment Tree
- **Subarray Sum Queries II** — Persistent Segment Tree / Mo's Algorithm
- **Distinct Values Queries** — Offline Queries + Fenwick Tree / Mo's Algorithm
- **Distinct Values Queries II** — Mo's Algorithm with Updates / Persistent Segment Tree
- **Increasing Array Queries** — Monotonic Stack / Segment Tree / Offline Queries
- **Movie Festival Queries** — Binary Lifting / Greedy
- **Forest Queries II** — 2D Fenwick Tree (BIT)
- **Range Updates and Sums** — Lazy Segment Tree
- **Polynomial Queries** — Lazy Segment Tree / Difference Arrays

- **Range Queries and Copies** — Persistent Segment Tree
- **Missing Coin Sum Queries** — Persistent Segment Tree / Merge Sort Tree

Tree Algorithms

- **Subordinates** — DFS / Tree DP
- **Tree Matching** — Tree DP / Greedy
- **Tree Diameter —** BFS/DFS twice / Tree DP
- **Tree Distances I** — Tree DP / Re-rooting DP
- **Tree Distances II —** Tree DP / Re-rooting DP
- **Company Queries I** — Binary Lifting (LCA setup)
- **Company Queries II —** Lowest Common Ancestor (Binary Lifting / RMQ / Euler Tour)
- **Distance Queries** — Lowest Common Ancestor
- **Counting Paths —** Lowest Common Ancestor / Tree Difference Array
- **Subtree Queries** — Euler Tour Technique / Fenwick Tree / Segment Tree
- **Path Queries —** Euler Tour Technique / Fenwick Tree
- **Path Queries II** — Heavy-Light Decomposition (HLD) / Segment Tree
- **Distinct Colors —** DSU on Trees (Sack) / Mo's Algorithm on Trees / Small-to-Large Merging
- **Finding a Centroid** — Tree DFS / Centroid Finding
- **Fixed-Length Paths I —** Centroid Decomposition
- **Fixed-Length Paths II** — Centroid Decomposition / Fenwick Tree

Mathematics

- **Josephus Queries** — Recursion / Math
- **Exponentiation** — Modular Exponentiation
- **Exponentiation II** — Modular Exponentiation / Fermat's Little Theorem
- **Counting Divisors** — Sieve of Eratosthenes / Prime Factorization
- **Common Divisors** — Array Counting / Divisor Iteration
- **Sum of Divisors** — Math / Arithmetic Progression / Modular Inverse
- **Divisor Analysis** — Math / Modular Exponentiation / Geometric Progression
- **Prime Multiples** — Inclusion-Exclusion Principle
- **Counting Coprime Pairs** — Inclusion-Exclusion Principle / Möbius Inversion
- **Next Prime** — Primality Testing (Miller-Rabin)

- **Binomial Coefficients** — Combinatorics / Modular Inverse / Factorials
- **Creating Strings II** — Combinatorics / Multinomial Coefficients
- **Distributing Apples** — Combinatorics / Stars and Bars
- **Christmas Party** — Combinatorics / Derangements
- **Permutation Order** — Combinatorics / Factoradic Base
- **Permutation Rounds** — Combinatorics / Eulerian Numbers
- **Bracket Sequences I** — Combinatorics / Catalan Numbers
- **Bracket Sequences II** — Combinatorics / Catalan Numbers variant
- **Counting Necklaces** — Burnside's Lemma / Combinatorics
- **Counting Grids** — Burnside's Lemma / Combinatorics
- **Fibonacci Numbers** — Matrix Exponentiation
- **Throwing Dice** — Matrix Exponentiation
- **Graph Paths I** — Matrix Exponentiation
- **Graph Paths II** — Matrix Exponentiation (Min-Plus Algebra)
- **System of Linear Equations** — Gaussian Elimination
- **Sum of Four Squares** — Number Theory / Legendre's Three-Square Theorem / Complete Search
- **Triangle Number Sums** — Math / Complete Search
- **Dice Probability —** DP / Probability
- **Moving Robots** — DP / Expected Value
- **Candy Lottery —** Probability / Expected Value / Math
- **Inversion Probability** — Expected Value / Linearity of Expectation / Fenwick Tree
- **Stick Game —** Game Theory DP / Nim variant
- **Nim Game I** — Game Theory / Nim (XOR Sum)
- **Nim Game II —** Game Theory / Nim (XOR Sum)
- **Stair Game** — Game Theory / Staircase Nim
- **Grundy's Game —** Game Theory / Sprague-Grundy Theorem
- **Another Game** — Game Theory / Nim

String Algorithms

- **Word Combinations** — DP + Trie / String Hashing

- **String Matching** — KMP Algorithm / String Hashing / Z-Algorithm
- **Finding Borders** — KMP Algorithm (Pi array) / Z-Algorithm / String Hashing
- **Finding Periods** — KMP Algorithm / Z-Algorithm / String Hashing
- **Minimal Rotation** — Booth's Algorithm / Suffix Array / String Hashing
- **Longest Palindrome** — Manacher's Algorithm / Binary Search + String Hashing
- **All Palindromes** — Manacher's Algorithm / Palindromic Tree
- **Required Substring** — KMP Automaton + DP
- **Palindrome Queries** — String Hashing + Segment Tree / Fenwick Tree
- **Finding Patterns** — Suffix Array / Aho-Corasick / Suffix Automaton
- **Counting Patterns** — Suffix Array / Aho-Corasick / Suffix Automaton
- **Pattern Positions** — Suffix Array / Suffix Automaton
- **Distinct Substrings** — Suffix Array / Suffix Automaton / LCP Array
- **Distinct Subsequences** — String DP
- **Repeating Substring** — Suffix Array / String Hashing + Binary Search
- **String Functions** — Z-Algorithm / KMP Algorithm
- **Inverse Suffix Array** — Suffix Array Inverse Construction
- **String Transform** — Burrows-Wheeler Transform (BWT) Inverse
- **Substring Order I** — Suffix Array + Segment Tree / Suffix Automaton
- **Substring Order II** — Suffix Array / Suffix Automaton
- **Substring Distribution** — Suffix Array + Difference Array / Suffix Automaton

Geometry

- **Point Location Test** — Cross Product
- **Line Segment Intersection** — Cross Product / Bounding Box Test
- **Polygon Area** — Shoelace Formula (Cross Product)
- **Point in Polygon** — Ray Casting Algorithm / Cross Product
- **Polygon Lattice Points** — Pick's Theorem / GCD
- **Minimum Euclidean Distance** — Divide and Conquer / Sweep Line
- **Convex Hull** — Graham Scan / Monotone Chain
- **Maximum Manhattan Distances** — Coordinate Transformation (x+y, x-y)
- **All Manhattan Distances** — Coordinate Transformation + Prefix Sums / Sorting

- **Intersection Points** — Sweep Line + Fenwick Tree
- **Line Segments Trace I** — Sweep Line
- **Line Segments Trace II** — Sweep Line / Segment Tree
- **Lines and Queries I** — Sweep Line / Geometry
- **Lines and Queries II** — Sweep Line / Geometry
- **Area of Rectangles** — Sweep Line + Segment Tree
- **Robot Path** — Sweep Line / Coordinate Compression + Set

Advanced Techniques

- **Meet in the Middle** — Meet in the Middle / Binary Search
- **Hamming Distance —** Bit Manipulation / GCC Pragmas
- **Corner Subgrid Check** — Bitsets
- **Corner Subgrid Count —** Bitsets
- **Reachable Nodes** — DAG / Bitsets / Topological Sort
- **Reachability Queries —** DAG / Bitsets / Topological Sort
- **Cut and Paste** — Treap / Implicit Treap / Splay Tree
- **Substring Reversals —** Implicit Treap / Splay Tree
- **Reversals and Sums** — Implicit Treap / Splay Tree
- **Necessary Roads —** Bridges / Tarjan's Bridge-Finding Algorithm
- **Necessary Cities** — Articulation Points / Tarjan's Algorithm
- **Eulerian Subgraphs —** DFS Tree / Back Edges Math
- **Monster Game I** — Convex Hull Trick (CHT)
- **Monster Game II** — Li Chao Tree / Dynamic Convex Hull Trick
- **Subarray Squares** — Divide and Conquer DP / Alien's Trick
- **Houses and Schools** — Divide and Conquer DP / Convex Hull Trick
- **Knuth Division** — Knuth Optimization for DP
- **Apples and Bananas** — Fast Fourier Transform (FFT) / Polynomial Multiplication
- **One Bit Positions** — Fast Fourier Transform (FFT)
- **Signal Processing** — Fast Fourier Transform (FFT)
- **New Roads Queries** — Kruskal Reconstruction Tree / Parallel Binary Search

- **Dynamic Connectivity** — Link-Cut Tree / Offline Dynamic Connectivity (Segment Tree over Time)
- **Parcel Delivery** — Min Cost Max Flow (MCMF)
- **Task Assignment** — Min Cost Max Flow (MCMF) / Hungarian Algorithm
- **Distinct Routes II** — Min Cost Max Flow (MCMF)

Sliding Window Problems

- **Sliding Window Sum** — Two Pointers / Sliding Window
- **Sliding Window Minimum —** Monotonic Queue / Deque
- **Sliding Window Xor** — Sliding Window / Trie
- **Sliding Window Or —** Sliding Window / Bitwise DP
- **Sliding Window Distinct Values** — Two Pointers / Hash Map
- **Sliding Window Mode** — Square Root Decomposition (Mo's Algorithm)
- **Sliding Window Mex** — Segment Tree / Mo's Algorithm
- **Sliding Window Median** — Two Heaps (Priority Queue) / PBDS / Segment Tree
- **Sliding Window Cost** — Two Heaps (Priority Queue) / PBDS / Median Finding Math
- **Sliding Window Inversions** — Fenwick Tree / PBDS / Square Root Decomposition
- **Sliding Window Advertisement** — Monotonic Stack / Segment Tree

Interactive Problems

- **Hidden Integer** — Binary Search (Interactive)
- **Hidden Permutation** — Interactive Logic
- **K-th Highest Score** — Interactive / Quickselect
- **Permuted Binary Strings —** Interactive Logic
- **Colored Chairs** — Interactive Setup
- **Inversion Sorting —** Interactive Sorting Algorithms

Bitwise Operations

- **Counting Bits** — DP / Math / Bit Manipulation
- **Maximum Xor Subarray** — Trie (Prefix XORs)
- **Maximum Xor Subset** — Gaussian Elimination over GF(2)
- **Number of Subset Xors** — DP / Subset Sum via FWHT (Fast Walsh-Hadamard Transform)

- **K Subset Xors** — DP / Fast Walsh-Hadamard Transform (FWT)
- **All Subarray Xors** — Prefix XORs / Math Counting
- **Xor Pyramid Peak** — Combinatorics (Lucas' Theorem) / Bitwise Math
- **Xor Pyramid Diagonal** — Combinatorics / Bitwise Math
- **Xor Pyramid Row** — Combinatorics / Lucas Theorem
- **SOS Bit Problem** — Sum Over Subsets (SOS) DP
- **And Subset Count** — SOS DP (Fast Zeta Transform)

Construction Problems

- **Inverse Inversions** — Constructive Algorithm / Math
- **Monotone Subsequences —** Constructive Algorithm / Erdős–Szekeres Theorem concept
- **Third Permutation** — Constructive Algorithm
- **Permutation Prime Sums —** Constructive Algorithm / Math
- **Chess Tournament** — Constructive Algorithm / Graph Theory
- **Distinct Sums Grid —** Constructive Algorithm
- **Filling Trominos** — Constructive Algorithm / Tiling
- **Grid Path Construction —** Constructive Algorithm / Backtracking

Advanced Graph Problems

- **Nearest Shops** — Multi-source BFS
- **Prüfer Code** — Tree Representation Math / Priority Queue
- **Tree Traversals** — Divide and Conquer
- **Course Schedule II** — Topological Sort / Priority Queue (Greedy)
- **Acyclic Graph Edges** — Topological Sort / DFS Backedges
- **Strongly Connected Edges** — DFS / Bridge Finding
- **Even Outdegree Edges** — DFS Tree
- **Graph Girth** — BFS (Shortest Cycle)
- **Fixed Length Walk Queries** — Matrix Exponentiation / Binary Lifting
- **Transfer Speeds Sum** — Maximum Flow
- **MST Edge Check** — Kruskal's Algorithm / Cycle Finding
- **MST Edge Set Check** — Kruskal's Algorithm / DSU
- **MST Edge Cost** — Kruskal's Algorithm / Cycle Analysis

- **Network Breakdown** — DSU (Reverse Operations)
- **Tree Coin Collecting I** — Re-rooting DP
- **Tree Coin Collecting II** — Re-rooting DP
- **Tree Isomorphism I** — Tree Hashing
- **Tree Isomorphism II** — Tree Hashing + Centroid Finding
- **Flight Route Requests** — DSU / Graph DP
- **Critical Cities** — Dominator Tree
- **Visiting Cities** — Shortest Path / Bridge Finding on DAG
- **Graph Coloring** — Bipartite Checking / Backtracking
- **Bus Companies** — DSU / Heavy-Light Decomposition (HLD)
- **Split into Two Paths** — Graph Theory / DFS
- **Network Renovation** — Tree Structure / Leaf Matching
- **Forbidden Cities** — Articulation Points / Biconnected Components / Block-Cut Tree
- **Creating Offices** — Tree Centroids / Greedy / BFS
- **New Flight Routes** — Strongly Connected Components / Indegree-Outdegree Analysis

Counting Problems

- **Filled Subgrid Count I** — Dynamic Programming / Math
- **Filled Subgrid Count II —** Dynamic Programming / Math
- **All Letter Subgrid Count I** — Combinatorics
- **All Letter Subgrid Count II —** Combinatorics
- **Border Subgrid Count I** — Math / Combinatorics
- **Border Subgrid Count II** — Math
- **Raab Game II** — Math / Game Theory
- **Empty String** — Dynamic Programming / Bracket Sequences Math
- **Permutation Inversions** — Dynamic Programming
- **Counting Bishops** — DP / Combinatorics
- **Counting Sequences** — Combinatorics / Generating Functions
- **Grid Paths II** — DP / Inclusion-Exclusion Principle
- **Counting Permutations** — Math / Combinatorics
- **Grid Completion** — Bitmask Dynamic Programming

- **Counting Reorders** — Combinatorics
- **Tournament Graph Distribution** — Math
- **Collecting Numbers Distribution** — Math
- **Functional Graph Distribution** — Math

Additional Problems I

- **Shortest Subsequence —** Greedy
- **Distinct Values Sum** — Math / Data Structures
- **Distinct Values Splits —** Dynamic Programming / Segment tree
- **Swap Game** — BFS / State Space Search
- **Beautiful Permutation II —** Backtracking / DP
- **Multiplication Table** — Binary Search
- **Bubble Sort Rounds I —** Math / Array Analysis
- **Bubble Sort Rounds II** — Math / Array Analysis
- **Nearest Campsites I —** 2D Segment Tree / KD Tree
- **Nearest Campsites II** — KD Tree / Sweepline
- **Advertisement —** Monotonic Stack
- **Special Substrings** — String Hashing / Hash Map
- **Counting LCM Arrays —** Math / Dynamic Programming
- **Square Subsets** — Math / Gaussian Elimination
- **Subarray Sum Constraints —** Sliding Window / Multiset
- **Water Containers Moves** — Graph / State Space
- **Water Containers Queries —** Trees / DSU
- **Stack Weights** — Segment Tree
- **Maximum Average Subarrays** — Binary Search / Prefix Sums
- **Subsets with Fixed Average** — DP / Meet in the Middle
- **Two Array Average** — Dynamic Programming
- **Pyramid Array** — Fenwick Tree
- **Permutation Subsequence** — Greedy / Math
- **Bit Inversions** — Segment Tree / Set
- **Writing Numbers** — Math / Digit DP

- **Letter Pair Move Game** — State Space Search
- **Maximum Building I** — Monotonic Stack (Max Area Rectangle)
- **Sorting Methods** — LIS / Greedy / Cycle Sort / Fenwick Tree
- **Cyclic Array** — Binary Lifting / Greedy
- **List of Sums** — Multiset / Math

Additional Problems II

- **Bouncing Ball Steps** — Math / Simulation
- **Bouncing Ball Cycle —** Math
- **Knight Moves Queries** — BFS / Math
- **K Subset Sums I —** Priority Queue
- **K Subset Sums II** — Priority Queue / DP
- **Increasing Array II —** Priority Queue (Slope Trick DP)
- **Food Division** — Median Finding / Math / Greedy
- **Swap Round Sorting —** Graph Theory / Cycle Analysis
- **Binary Subsequences** — Greedy / Math
- **School Excursion —** DP / Knapsack (Bitset optimization)
- **Coin Grid** — Bipartite Matching (Kőnig's Theorem) / Max Flow
- **Grid Coloring II —** Math / Constructive Algorithm
- **Programmers and Artists** — Greedy / Sorting / Priority Queue
- **Removing Digits II —** DP / Recursion
- **Coin Arrangement** — Greedy / Math
- **Replace with Difference** — Math
- **Grid Puzzle I** — Bipartite Matching / Max Flow
- **Grid Puzzle II** — Min Cost Max Flow (MCMF)
- **Bit Substrings** — String / Math / DP
- **Reversal Sorting** — Implicit Treap / Splay Tree
- **Book Shop II** — Knapsack DP (Bounded Knapsack Optimization)
- **GCD Subsets** — Math / DP
- **Minimum Cost Pairs** — Dynamic Programming / Graph
- **Same Sum Subsets** — DP / Meet in the Middle

- **Mex Grid Queries** — Segment Tree / Mo's Algorithm
- **Maximum Building II** — Monotonic Stack / Difference Arrays
- **Stick Divisions** — Priority Queue (Huffman Coding concept)
- **Stick Difference** — Set / Multiset